Sandip Subhrajit Sahoo

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.10
Upendra Nath H.S. School, Soro	Class XII (CHSE Odisha)	2023	88.50%
Kanchan Kamala High School, Ramakrushnapur	Class X (BSE Odisha)	2021	88.67%

SCHOLASTIC ACHIEVEMENTS

Rank 5, Optiver Trade-a-thon 2025, IITH: Competed in a live-trading simulation, utilizing mathematical expressions and rapid mental math to predict future asset prices, execute profitable trades, and manage risk against other participants.

Qualified Stage 2 of **AlgoUniversity Technology Fellowship 2025**, ranking in the **top 8%** among 2,50,000+ participants.

PROJECTS

AI Video Assistant with RAG | GitHub Link

Tech: Python, FastAPI, LLM, LangChain, RAG, Tensorflow

- Engineered a production-style **video understanding system** that ingests YouTube links and multimedia files, performs transcription, semantic indexing, and conversational retrieval over video content.
- Designed and implemented a **Retrieval-Augmented Generation (RAG) architecture** combining vector embeddings, similarity search, and LLM-based response generation for grounded question answering.
- Built scalable document-processing pipelines for transcript chunking, embedding generation, metadata management, and contextual retrieval from long-duration recordings.
- Enhanced user productivity through automated extraction of summaries, key decisions, and action items, reducing manual review effort for meetings and educational content.

Multi-Agent LLM Framework | GitHub Link

Tech: Python, LangGraph, LangChain, LLMs, RAG, Vector Search, Prompt Engineering

- Built a production-style **multi-agent LLM framework** where specialized agents collaborated through structured workflows to perform planning, information retrieval, reasoning, execution, and response verification.
- Designed an intelligent **agent orchestration layer** supporting task delegation, inter-agent communication, memory management, and state-aware workflow execution.
- Implemented RAG-based knowledge retrieval using embeddings and vector databases, enabling agents to access external information and generate grounded responses.
- Enhanced reliability through reviewer and validation agents that evaluated intermediate outputs, improving response quality and reducing reasoning errors in multi-step tasks.

NanoGPT: GPT-based Language Model from Scratch| Github Link

Tech: Python, PyTorch, NLP, Transformers, Deep Learning

- Built a **GPT-style Large Language Model (LLM) from first principles**, implementing the complete Transformer decoder architecture with masked self-attention and autoregressive text generation.
- Designed a scalable training framework supporting tokenization, data loading, distributed mini-batch training, checkpoint management, and inference-time sampling strategies.
- Implemented and optimized core transformer components including **multi-head attention, positional embeddings, residual pathways, layer normalization, and feed-forward blocks**, achieving stable convergence during training.
- Reproduced foundational techniques used in modern generative AI systems, providing practical understanding of LLM pretraining, sequence modeling, and Transformer-based language generation.

SKILLS

Programming Languages: C++, Python, C, Objective-C++

Systems & Low-Latency: Unix/POSIX APIs (kqueue, libproc), Memory Management(Zero-Heap Allocation, Cache Locality), Multithreading, Event-Driven Architecture, Operating Systems Scheduling

AI & Scientific Computing: PyTorch, Scikitlearn, TensorFlow, Keras, NumPy, SciPy, Pandas, Custom Inference Architectures **Developer Tooling:** Git, CMake, LLDB/GDB (or System Profilers), Shell Scripting, Linux/Darwin Environments

RELEVANT COURSES

Mathematics: Calculus-I, Calculus- II, Elementary Linear Algebra, Differential Equations, Transform Techniques, Introduction to Probability, Introduction to Statistics, Complex Variables.

Computer Science: Data Structures and Applications, Operating System-II, Introduction to Programming

Computational Methods: Fundamentals of Scientific Computing, Computer-Aided Numerical Methods I, Computer-Aided Numerical Methods II, Computational Methods in Material Sciences, Electronic Structure & Atomistic Modelling of Materials.

POSITIONS OF RESPONSIBILITY

Co-Ordinator

Content and speaker curator

EXTRACURRICULAR

Participated in multiple District-level chess tournaments.

Participated in regional Cricket tournaments